

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Atmakuri Sai Mrudula
Register Number	192372087

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I **Atmakuri Sai Mrudula (192372087)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **A. Sai Mrudula**

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Library Book Reservation System

Aim:

To develop a cloud-based Library Book Reservation System for managing books and student reservations.

Procedure:

1. Create Student, Book Details, and Reservation forms.
2. Add book information and availability.
3. Connect Student and Book forms using Lookup fields.
4. Record issue and return dates.
5. Generate reservation reports.
6. Test the reservation process.
7. Publish the application.

Implementation:

The screenshot displays the 'Library Book Reservation System' web application. The interface features a dark blue sidebar on the left with navigation links: 'Customer', 'Customer Report', 'Student', 'Book Details', and 'Reservation'. The main content area is titled 'Customer' and contains a form for adding a new customer. The form includes fields for 'Name' (split into 'First Name' with the value 'Mrudula' and 'Last Name' with the value 'Sai') and 'Phone' (with a dropdown for country code '+91' and a text input for the number '987456321'). A blue 'Submit' button is positioned below the form fields. The top of the application shows a header with the system name, user avatars, and links for 'Trial expires in 1 day', 'Upgrade', 'Edit this application', and 'Help'. The bottom of the image shows a Windows taskbar with various icons and the system clock indicating 10:50 PM on 10/10/2023.

Library Book Reservation System

Library Book Reservation System

Customer

Customer

Customer Report

Student

Book Details

Reservation

saimrudulaa2...

Customer Report

Name

Phone

Mrudula Sai

+919874563210

Showing 1 of 1

Library Book Reservation System

Library Book Reservation System

Customer

Student

Student

Student Report

Book Details

Reservation

saimrudulaa2...

Student

Student ID

1001

Student Name

Sai

First Name

Last Name

Department

CSE

First Name

Last Name

Year

2

Phone

+91

8745693210

Submit

30°C

Search

ENG

10:51 PM

Library Book Reservation System

Library Book Reservation System

Customer >Student >Student >Student Report >Book Details >Reservation >

Showing 1 of 1

Student Report

Student ID	Student Name	Department	Year	Phone
1001	Sai	CSE	2	+918745693210

30°CPartly cloudy

Library Book Reservation System

Library Book Reservation System

Customer >Student >Book Details >Book Details >Book Details Report >Reservation >

Book Details

Book ID101

TitleThe princess September

Author

Anil

Kumar

First NameLast Name

Publication Year2021

Number of Copies100

StatusPublished

Submit

30°CPartly cloudy

Library Book Reservation System

Customer >Student >Book Details >Book Details Book Details Report >Reservation >

saimrudulaa2...

Book Details Report

Book ID	Title	Author	Publication Year	Number of Copies	Status
101	The princess September	Anil Kumar	2021	100	Published

Showing 1 of 1

30°C
Partly cloudy

Search

10:54 PM
7/25/2026

Library Book Reservation System

Customer >Student >Book Details >Reservation >Reservation Report

saimrudulaa2...

Reservation

Student

Sai

Book Details

The princess September

Issue Date

26-Jul-2026

Return Date

29-Jul-2026

Fine

0

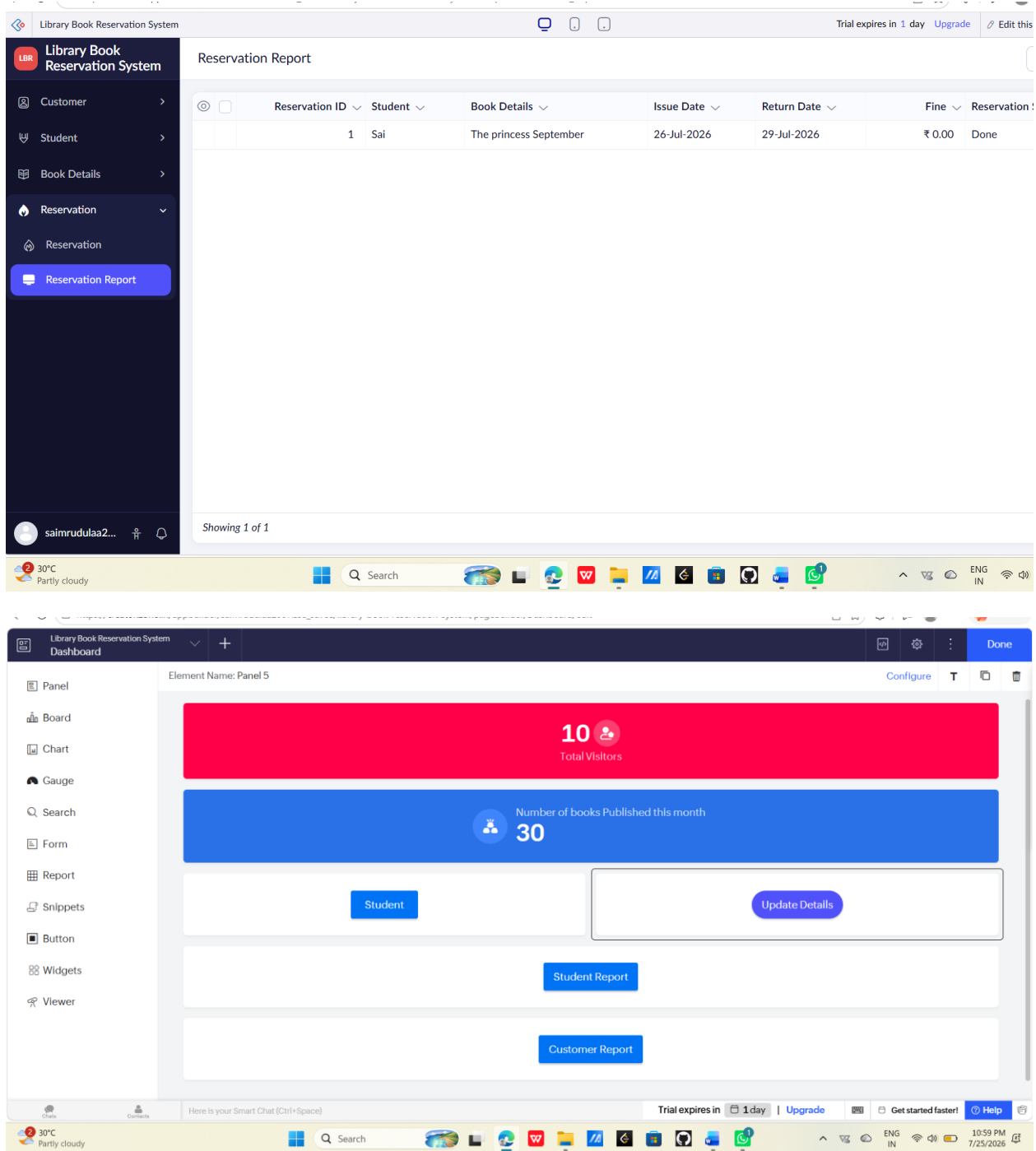
Reservation Status

Done

Submit

30°C

Search



Result:

A cloud-based Library Book Reservation System was successfully developed. The application manages book inventories, reservations, issue/return details, and book availability efficiently.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Payroll Processing System

Aim:

To develop a cloud-based Payroll Processing System for managing employee salary information.

Procedure:

1. Create Employee, Salary Structure, and Payroll forms.
2. Enter employee information.
3. Add salary components such as Basic Pay, HRA, DA, TA, Bonus, PF, and Tax.
4. Connect Employee and Salary forms using Lookup fields.
5. Calculate Gross Pay and Net Pay.
6. Generate payroll reports.
7. Publish the application.

Implementation:

The screenshot displays a web application titled "Payroll Processing System" in a browser window. The address bar shows the URL: https://creatorapp.zoho.in/saimrudulaa2087.sse_savee/payroll-processing-system1#Formdeductions. The application has a dark sidebar menu on the left with options: Dashboard, Payroll Records, Deductions (highlighted with a red plus icon), Deductions Report, Board, Employees, Payroll Disbursement, Users, Payroll Validations, Employee Salaries, and Pay Periods. The main content area is titled "Deductions" and contains a form with the following fields: Employee (dropdown menu), Pay Period (dropdown menu), Deduction Type (dropdown menu), Amount (text input with a mask #####.##), Percentage (text input), Effective Date (date input in dd-MMM-yyyy format), End Date (date input in dd-MMM-yyyy format), Is Active (checkbox), and Notes (text area). At the bottom of the form are "Submit" and "Reset" buttons. The top right of the application shows a notification "Trial has expired" with an "Upgrade" button, and links for "Edit this application" and "Help".

Payroll Processing System

Dashboard

Payroll Records

Deductions

Deductions Report

Board

Employees

Payroll Disbursement

Users

Payroll Validations

Employee Salaries

Pay Periods

Payroll Processing System

Deductions Report

Employee	Pay Period	Deduction Type	Amount	Percentage	Effective Date	End Date	Is Active	Notes
Elena	August 2026 June 2026	Health Insurance	\$ 50.00	1.50	01-Nov-2026	31-Dec-2026	true	Annual retirement
Elena	July 2026 June 2026 May 2026	Health Insurance	\$ 100.00	0.00	01-Oct-2026	31-Oct-2026	false	Reimbursement
Alice	July 2026 June 2026	401k	\$ 75.50	2.50	01-Sep-2026	30-Sep-2026	true	Quarterly insurance
Elena	August 2026	Other	\$ 200.00	10.00	15-Aug-2026	14-Aug-2026	true	One time bonus
Daniel	July 2026 June 2026 May 2026	Union Dues	\$ 150.00	5.00	01-Jul-2026	31-Dec-2026	true	Monthly tax deduction

Showing 5 of 5

Show Summary

Payroll Processing System

Dashboard

Payroll Records

Deductions

Employees

Payroll Disbursement

Users

Payroll Validations

Employee Salaries

Employee Salaries

Board

Pay Periods

Payroll Processing System

Employee Salaries

Employee

Effective Date

Base Salary

Hourly Rate

Overtime Rate

Salary Type

Pay Frequency

Currency

Notes

Payroll Processing System

Employee Salaries Report

Employee	Effective Date	Base Salary	Hourly Rate	Overtime Rate	Salary Type	Pay Frequency	Currency	Notes
Carol	01-Sep-2026	\$ 50,000.00	\$ 25.00	\$ 37.50	Commission	Monthly	GBP	Con
Alice	27-Jul-2026	\$ 75,000.00	\$ 37.50	\$ 56.25	Fixed	Bi-Weekly	USD	Bon
Alice	15-Jul-2026	\$ 48,000.00	\$ 24.00	\$ 36.00	Hourly	Bi-Weekly	EUR	New
Robert	01-Aug-2026	\$ 62,000.00	\$ 31.00	\$ 46.50	Hourly	Bi-Weekly	USD	Pror
Carol	27-Jul-2026	\$ 55,000.00	\$ 27.50	\$ 41.25	Fixed	Monthly	USD	Ann

Showing 5 of 5

Show Summary

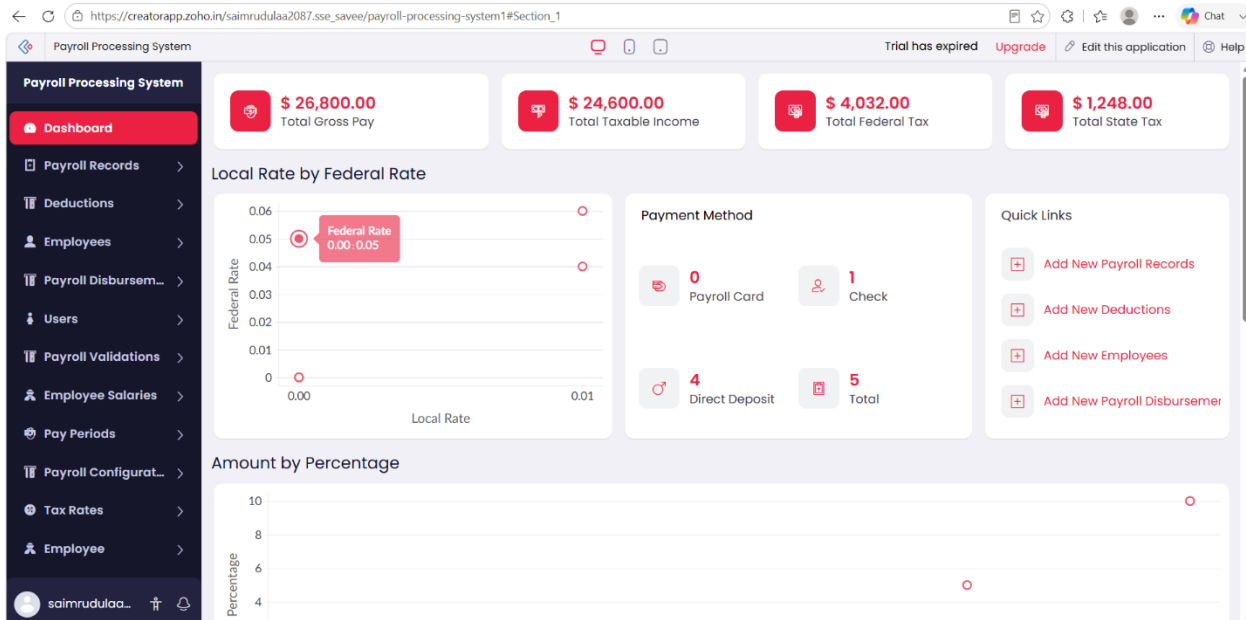
Payroll Processing System

Payroll Configurations Report

Company Name	Tax Filing Status	Payroll Frequency	Auto Approve	Default Tax Rate	Default Deduction	Created By	Created
	Head of Household	Weekly	false				
	Head of Household	Monthly	false				
	Single	Weekly	false				

Showing 3 of 3

Show Summary



Result:

A cloud-based Payroll Processing System was successfully developed. The application calculates employee salaries accurately and generates payroll reports.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

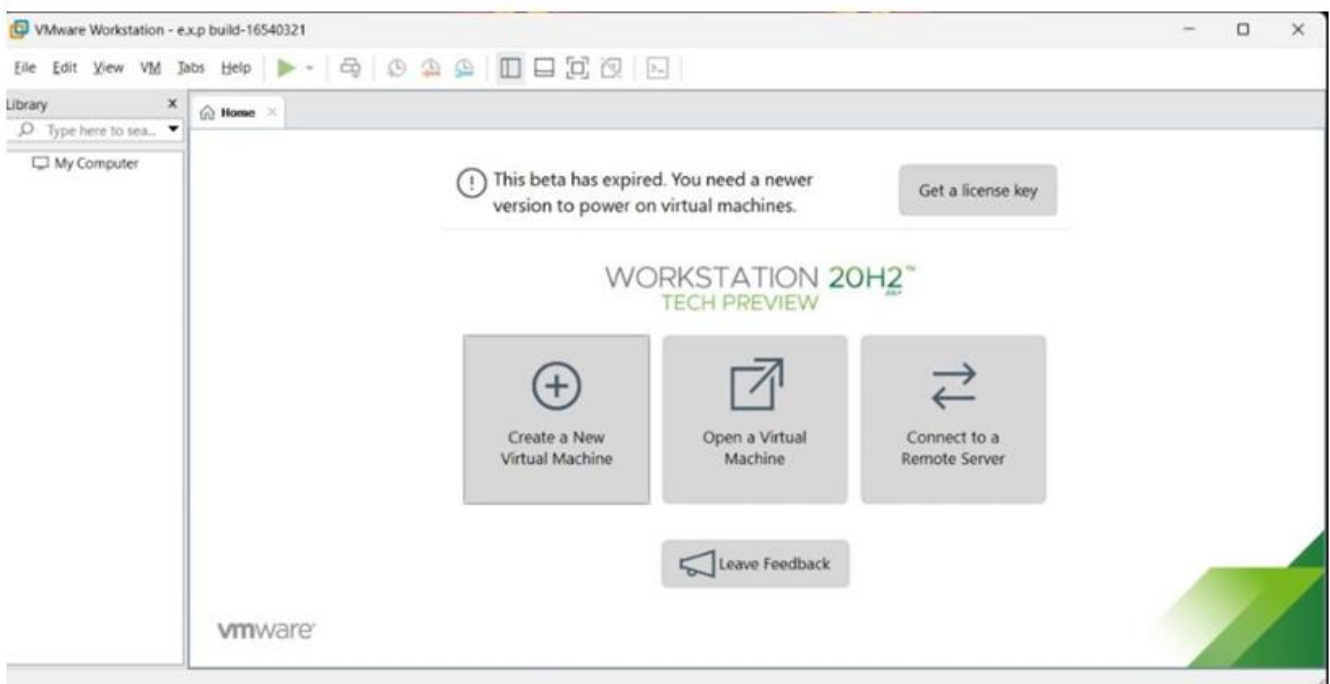
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage.

Procedure:

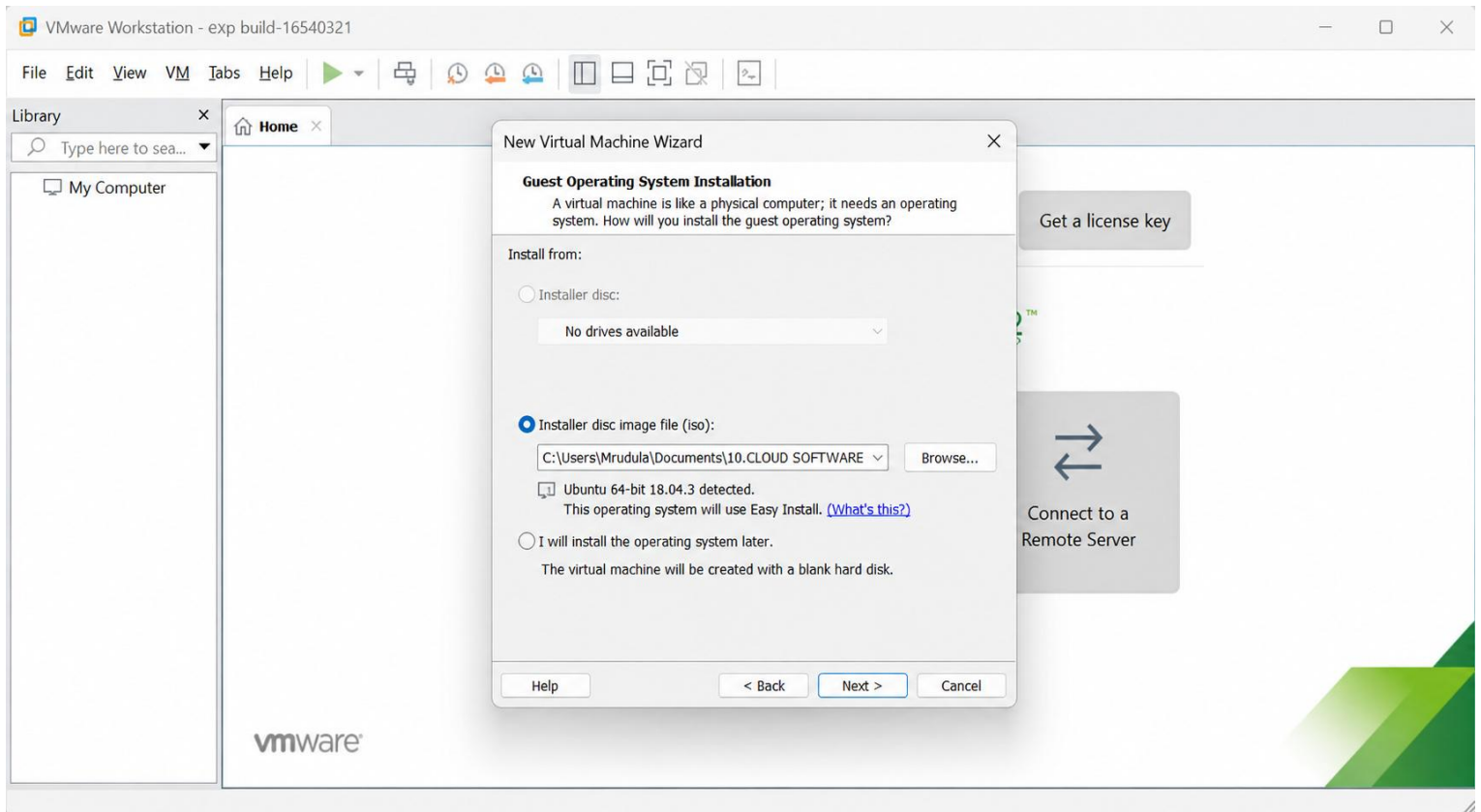
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

Implementation

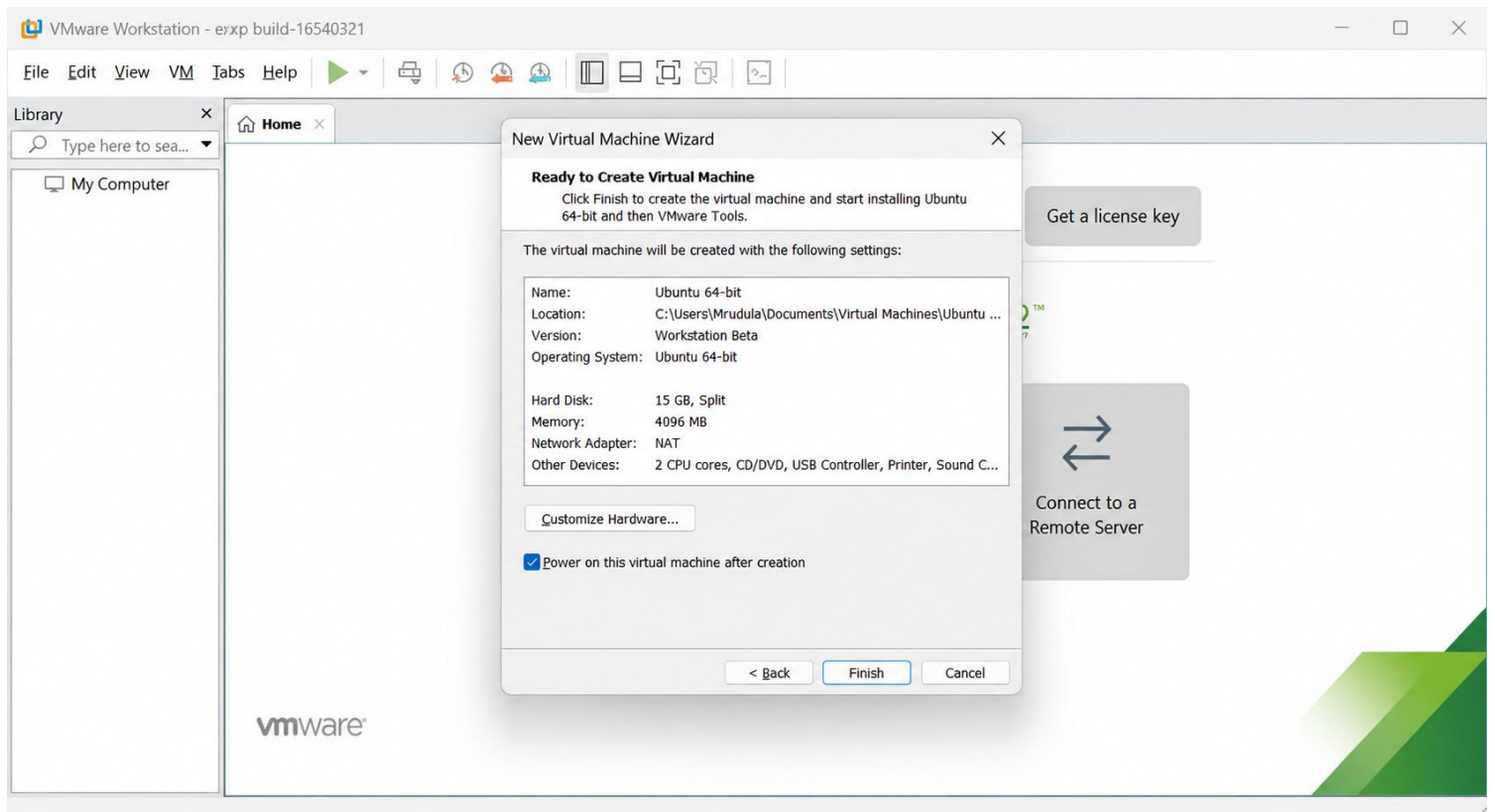
1. Open VMware and select Create New Virtual Machine.



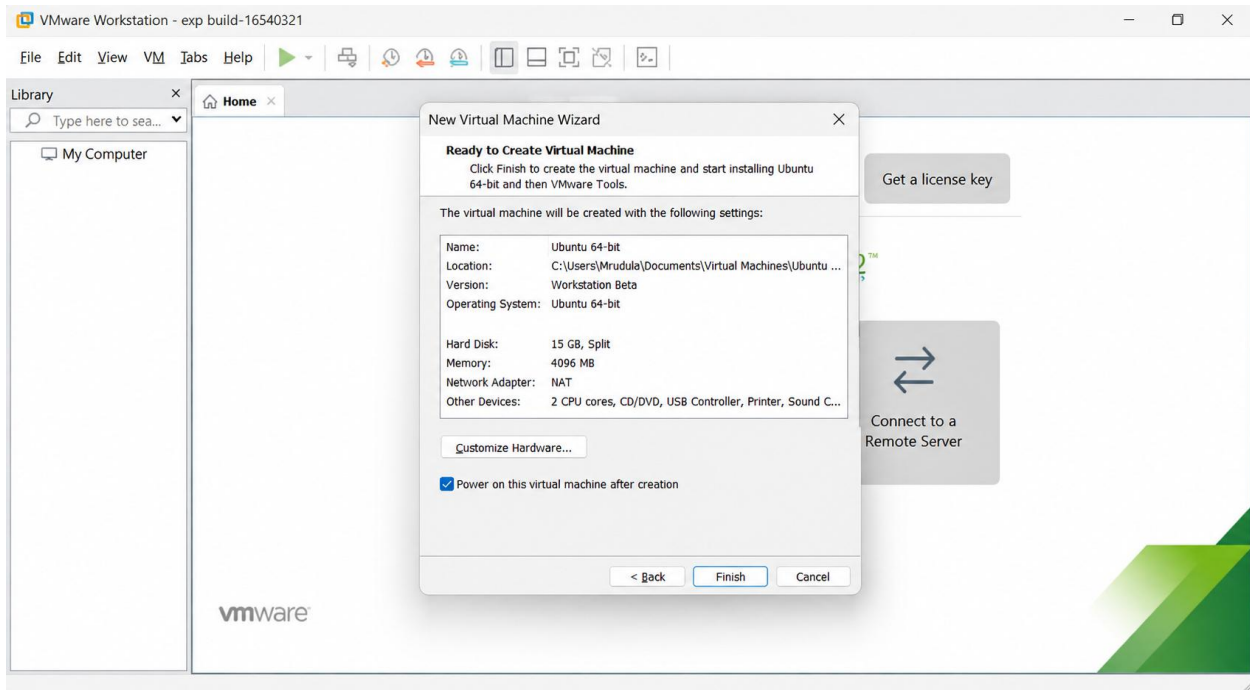
2. Select the Ubuntu ISO file.



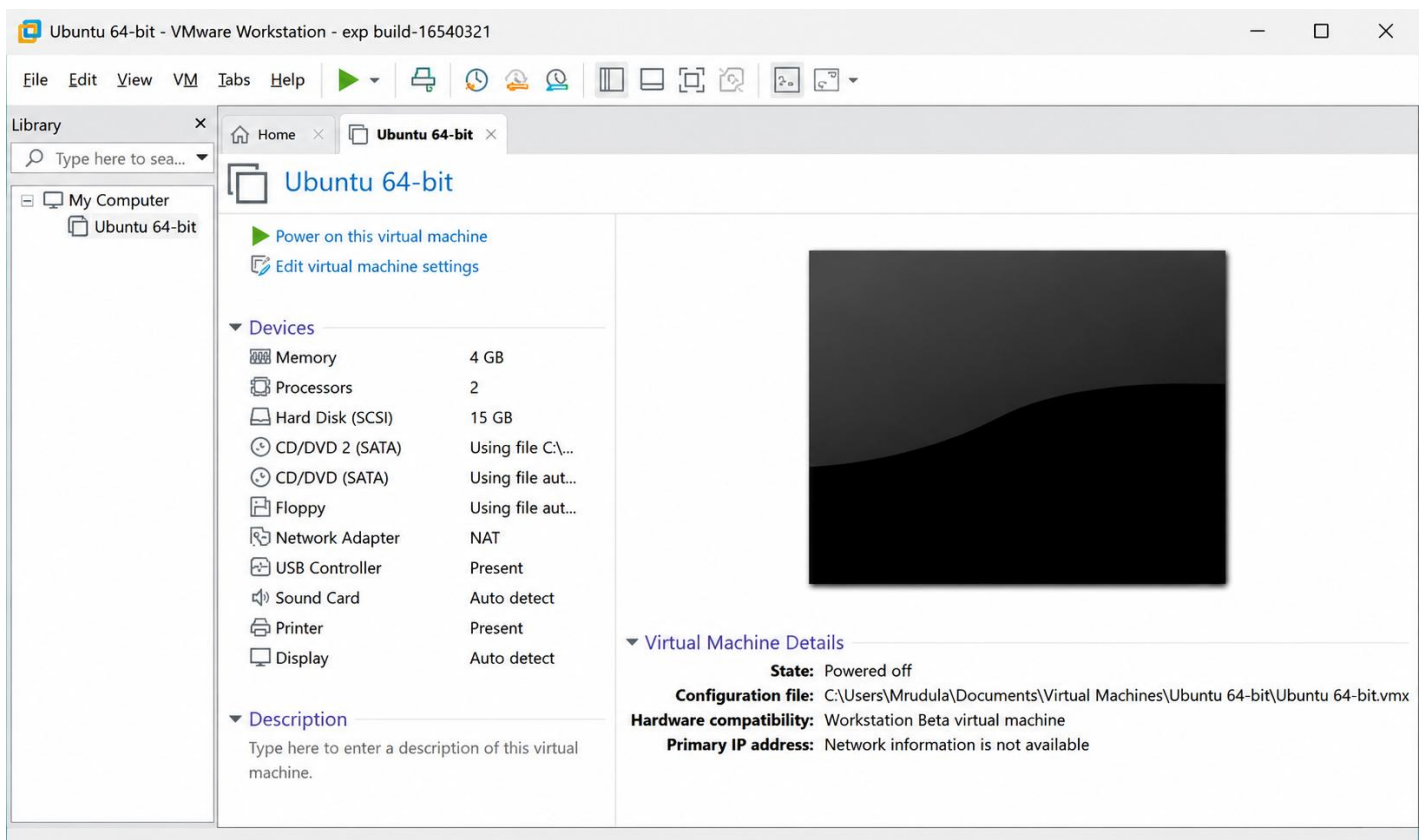
3. Set the VM name as Ubuntu-VM.



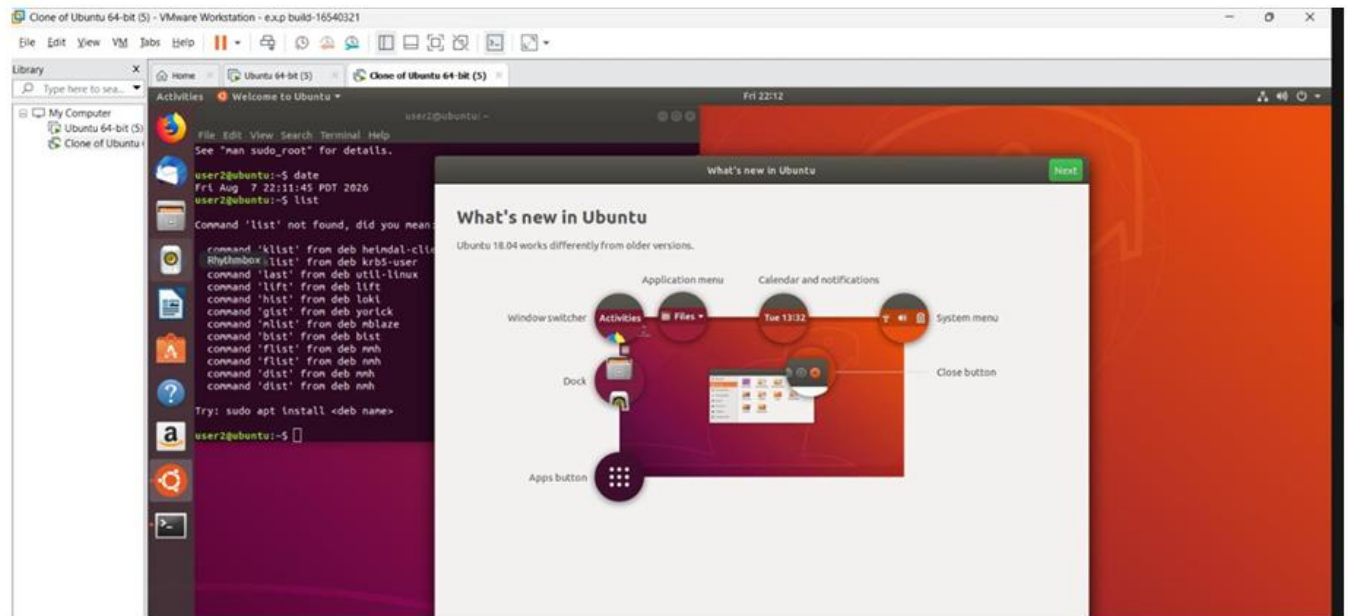
4. Set RAM = 2 GB and CPU = 1.



5. Set the hard disk size = 15 GB



6.Start the VM and install Ubuntu.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim: -

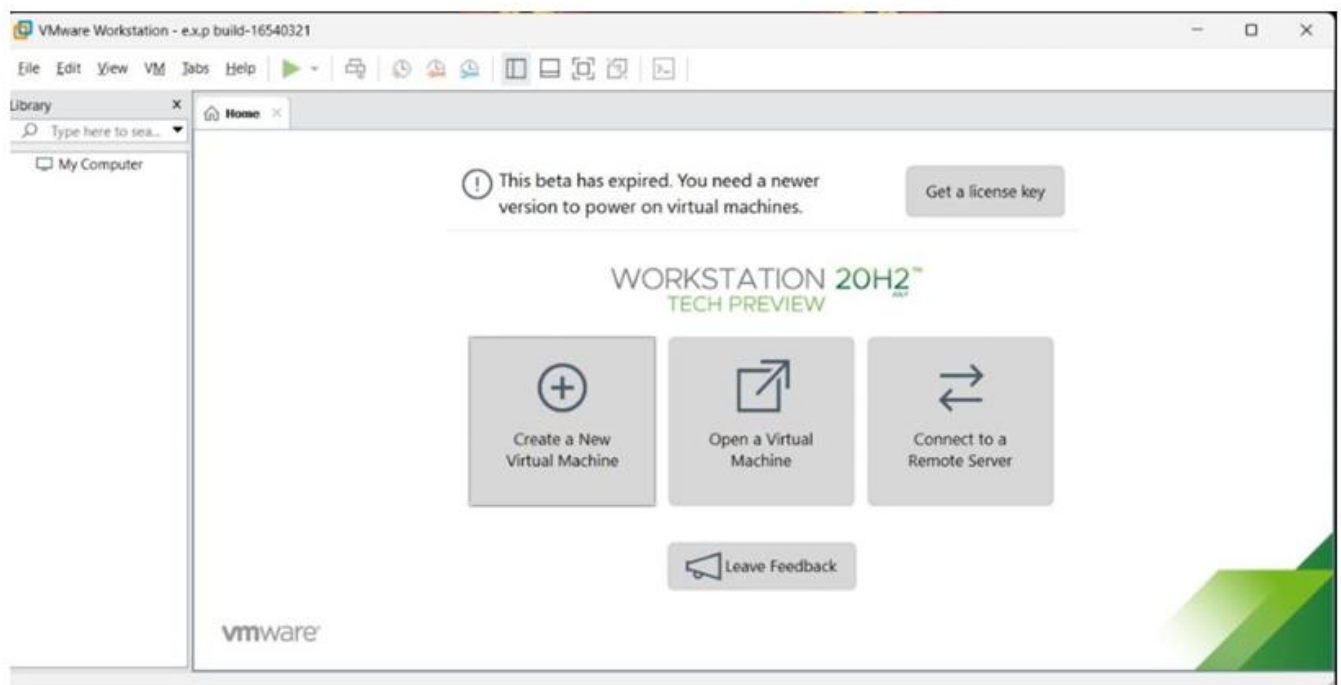
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation), creating a Virtual Machine with a Windows/Linux operating system, and cloning the VM to create and test a previous version of the Virtual Machine.

Procedure: -

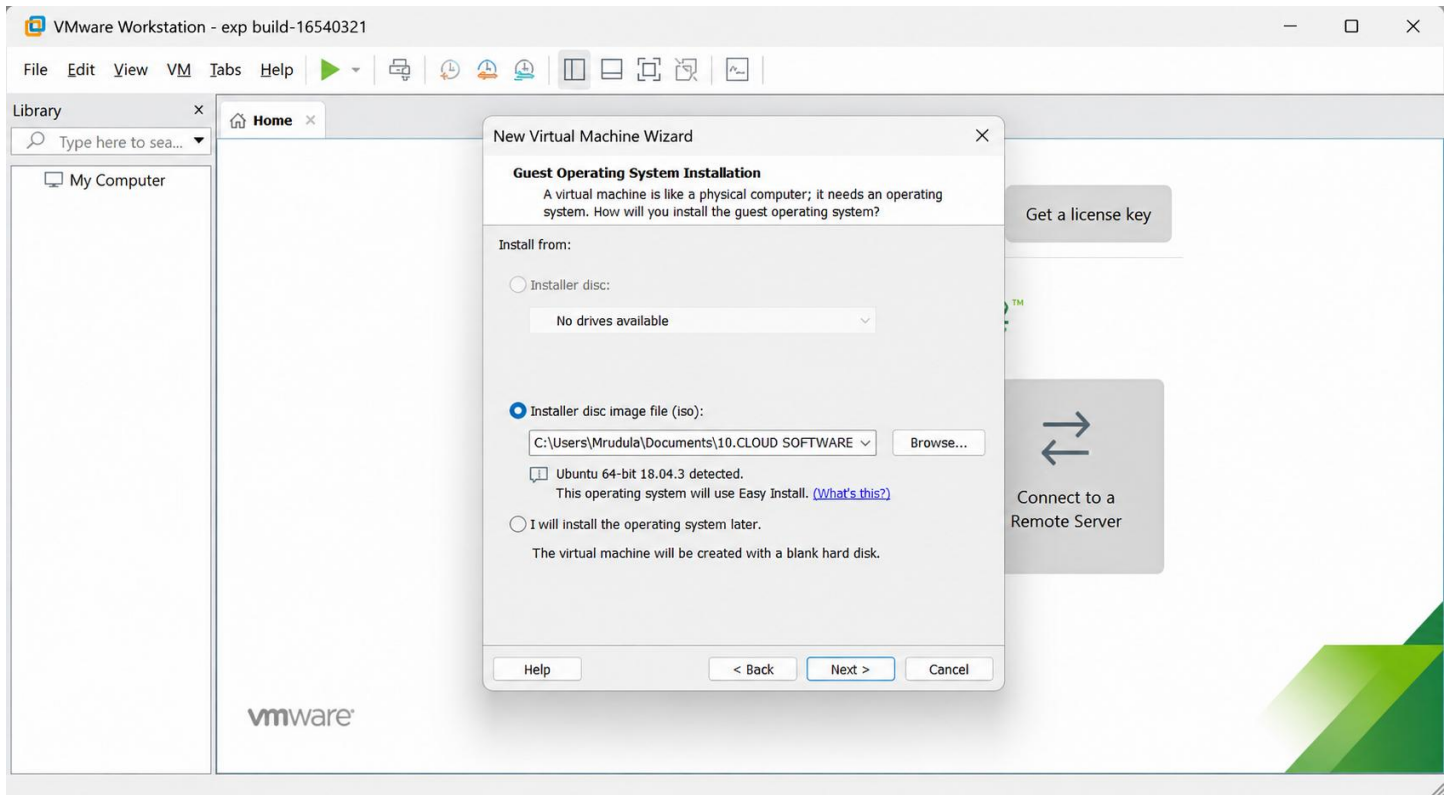
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu

Implementation: -

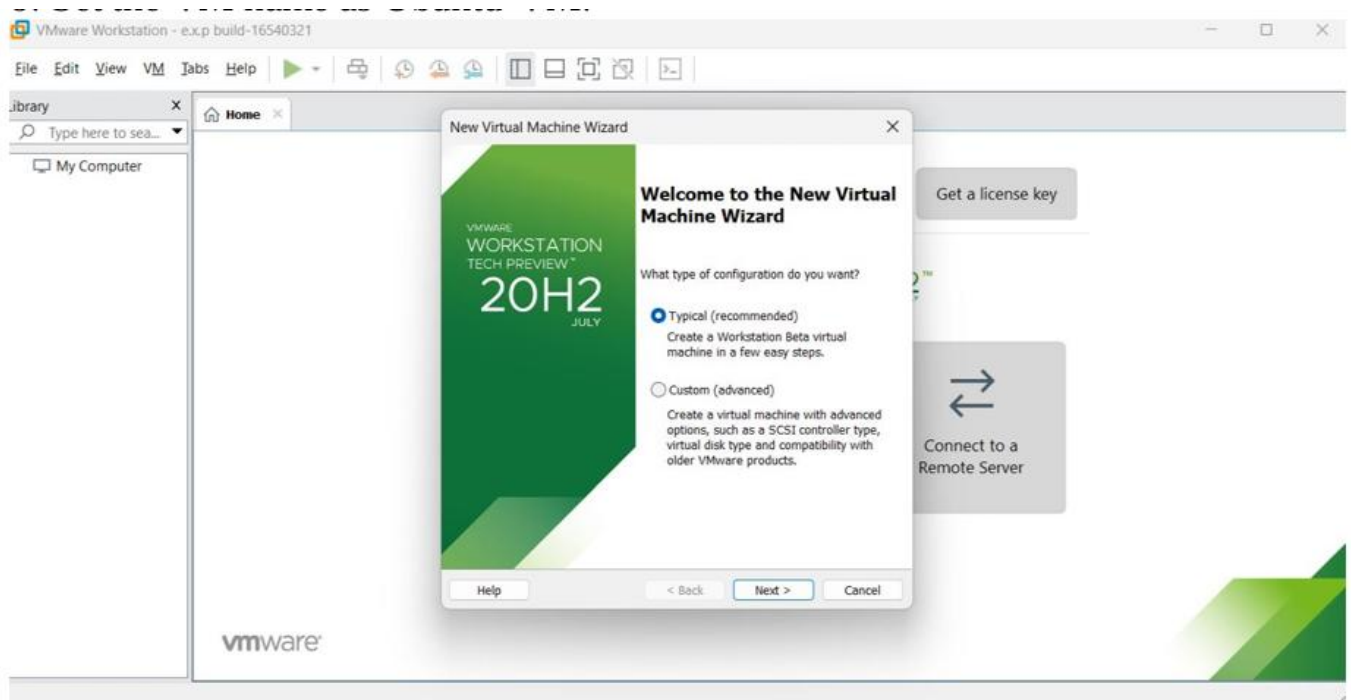
1. Open VMware and select Create New Virtual Machine.



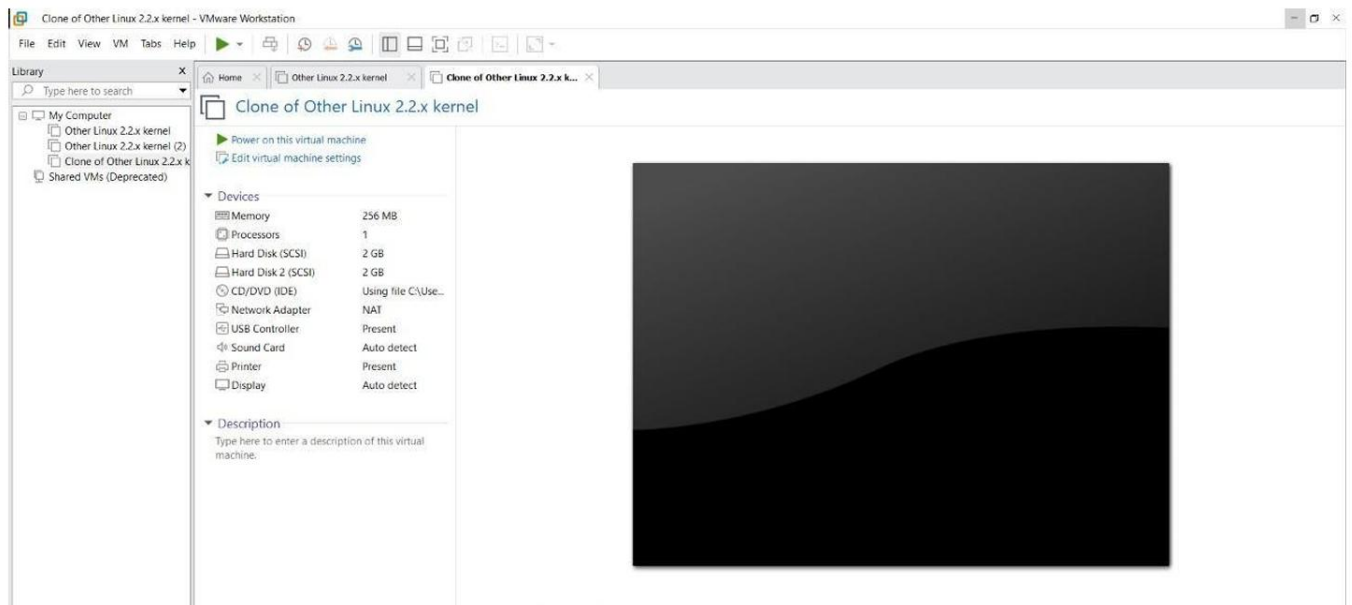
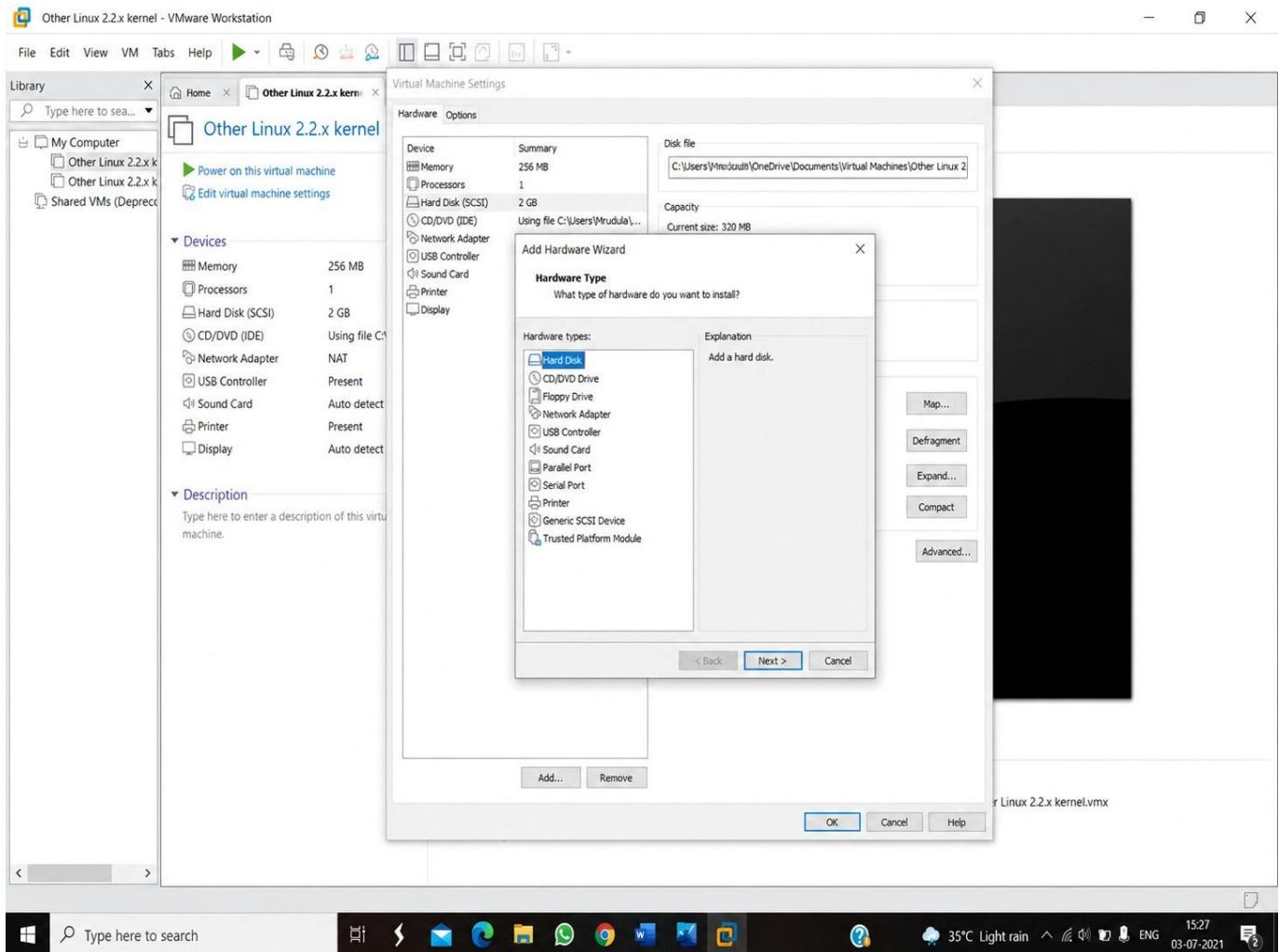
2. Select the Ubuntu ISO file.

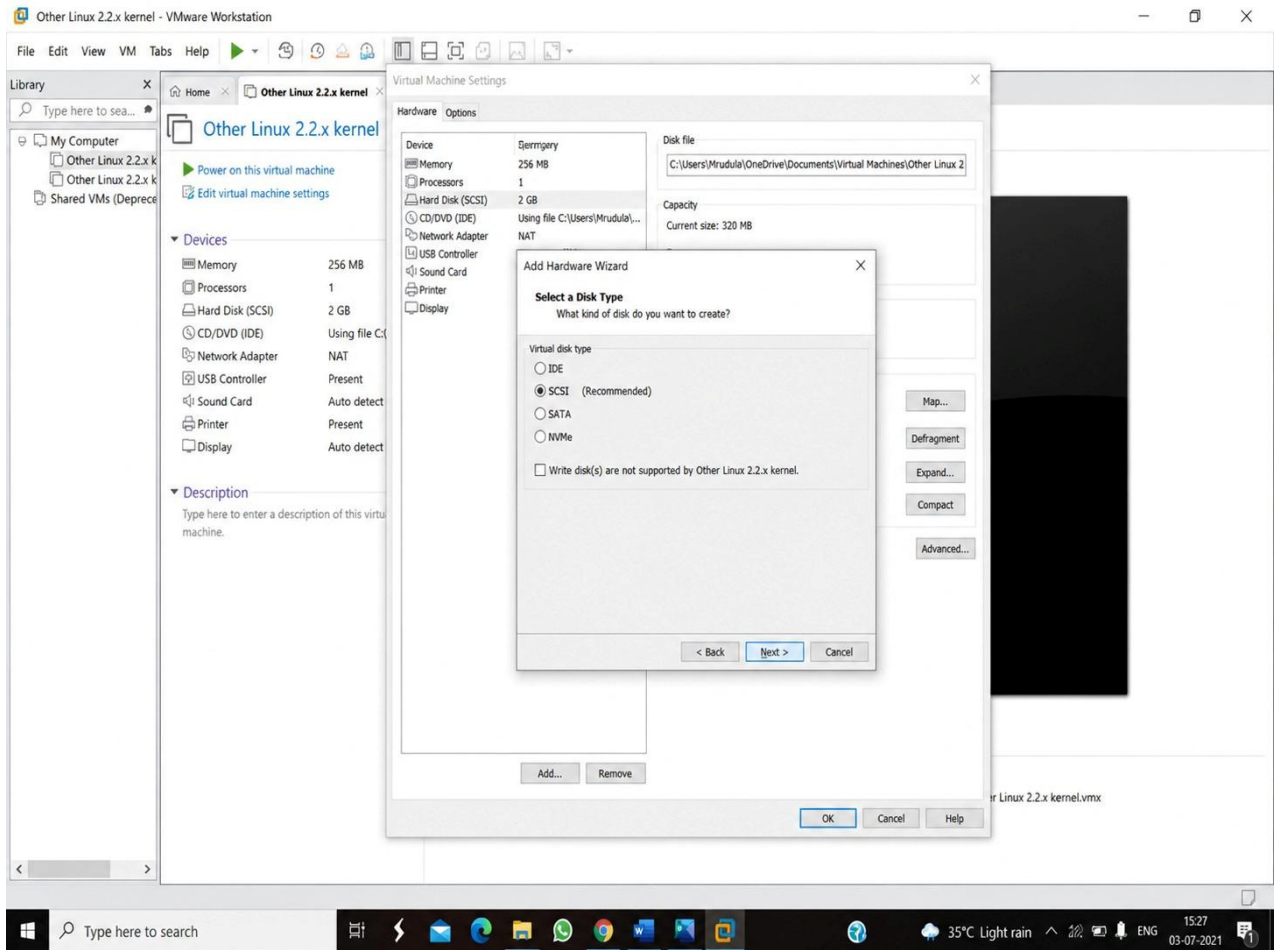


3. Set the VM name as Ubuntu-VM.

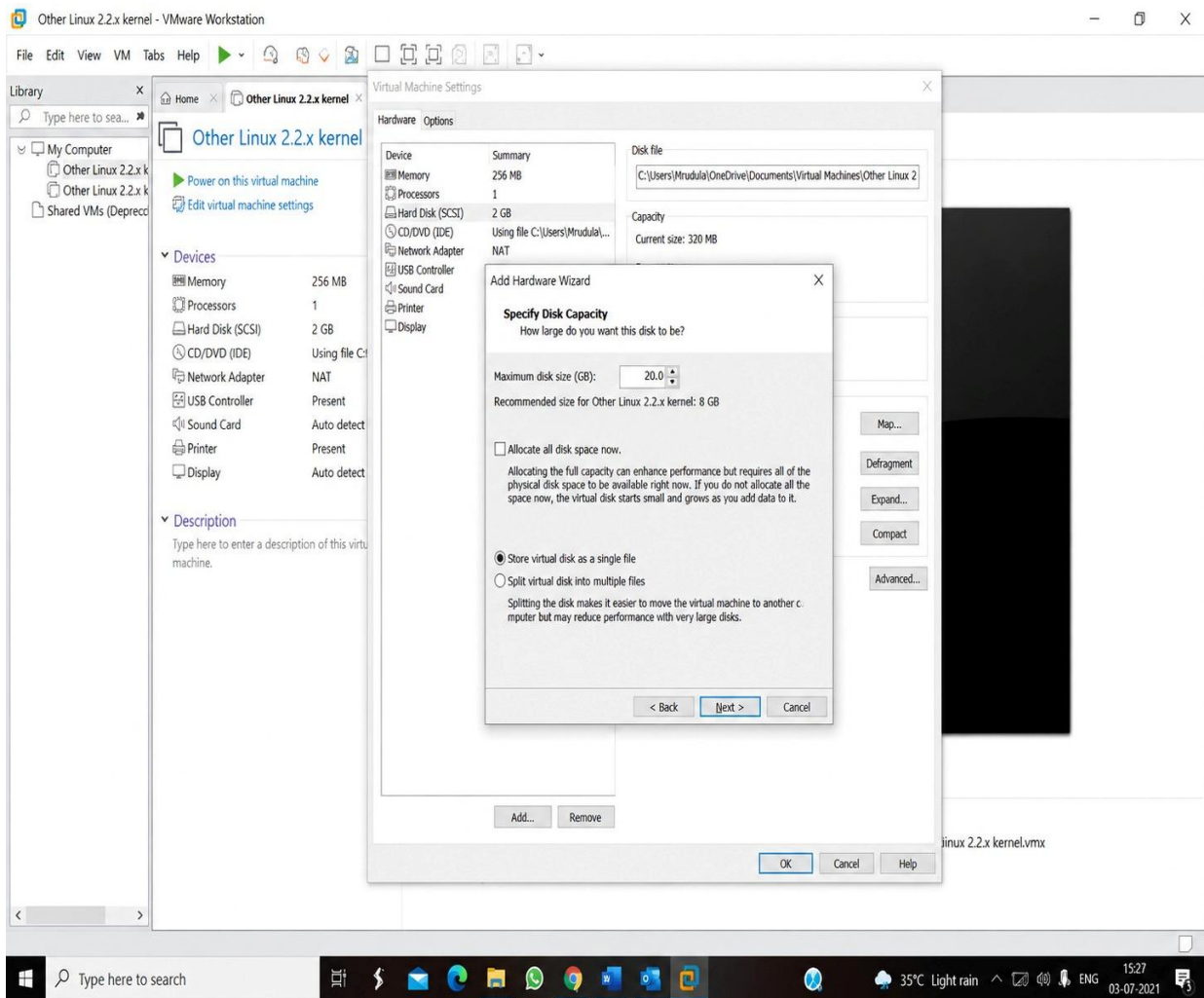


4. Set RAM = 2 GB and CPU = 1.





STEP 5: SELECT THE DISK SIZE AS 2.0. AND SELECT SPLIT VIRTUAL DISK INTO MULTIPLE FILES.



STEP 6: GIVE NAME AND CLICK THE FINISH

